Gunturu Madhukar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.16
DR KKR GOWTHAM SCHOOL, TENALI, GUNTUR DISTRICT	Class XII (Board of Secondary Education, Andhra Pradesh)	2023	90.20%
Sri Chaitanya Junior College, KANURU, VIJAYAWADA	Class X (Board of Intermediate Education, Andhra Pradesh)	2021	100.00%

SKILLS

Programming Languages: C, C++, Python

Tools: Linux, Git, LaTeX

RELEVANT COURSES

CS related courses: Data Structures and Applications, Introduction to AI, Data Science Analytics, Machine Learning for Physical Sciences, Machine Learning for High Energy Physics, Computational Physics, Introduction to Programming

Mathematics & EE: Introduction to Probability, Introduction to Statistics, Elementary Linear Algebra, Calculus, Linear Systems and Signal Processing, Digital Circuits, Analog Electronics, CMOS Processing

PROJECTS

Robust Pupil Detection Using Classical Computer Vision and Deep Learning | (B.Tech Project under Prof. Vandana Sharma)

- Developed a **YOLOv8n-based pupil detection pipeline** for noisy eye videos, replacing the earlier **U-Net image reconstruction** approach.
- Deployed the trained model in **NCNN format on Raspberry Pi** with **real-time pupil-area tracking, fatigue analysis and GUI-based visualization**.

Neural Network-Assisted Variational Quantum Circuit Pipeline | (Course Work Prof. Sangkha Borah)

- Developed a **hybrid quantum-classical VQE pipeline** using **neural networks for parameter initialization** in deep variational quantum circuits.
- Implemented **gradient-variance benchmarking** and **NN-based parameter generation** for a **6-qubit Transverse-Field Ising Model**.
- Built **Hamiltonian-transfer benchmarking, candidate parameter ranking and multi-format model export** pipelines.

Benford's Law Applied to the NASA Exoplanet Archive | (Course Work Prof. Shantanu Desai)

- Analysed physical parameters of confirmed exoplanets from the **NASA Exoplanet Archive** to test conformity with **Benford's Law**.
- Implemented **Chi-squared, Kullback-Leibler divergence, Kolmogorov-Smirnov and Bayesian statistical tests** across **seven parameters**.
- Investigated **parameter-wise deviations** and variation across **temporal and exoplanet discovery-method datasets**.

Common Ground: Bounded Shared Knowledge in Decentralised Ambulance Dispatch | (Course Work Prof. Venkatraman Renganathan)

- Implemented a **multi-agent ambulance-dispatch simulator** on the **Hyderabad road network** under **range-limited and unreliable communication**.

- Developed the **Hyderabad road graph, zone-based travel-time cache, event-driven simulation engine and dashboard framework**.
- Implemented **distributed redeployment** and **bounded-depth dispatch protocols (P0/P1/P2)** with **response-time, coverage and duplicate-dispatch metrics**.

EXTRACURRICULAR

- Badminton and Football enthusiast